Practical 2, tutorial question: when does an iteration actually converge?

MTH2014 Numerical Modelling – Week 2. Pen and paper, in pairs if you like! No computer needed for this one – a calculator is enough.

Part 1: fixed-point iteration

Recall the fixed-point iteration $x_{n+1} = g(x_n)$.

(a) Take $g(x) = \cos(x)$ (radians) and $x_0 = 0.2$. Calculate $x_1, \ldots, x_5$ by hand, to 5 decimal places.

(b) Now take $g(x) = \cos^{-1}(x)$, with the same $x_0 = 0.2$. Calculate as many iterations as you can go before something goes wrong. What happens, exactly?

(c) Both of these are trying to converge to the same fixed point (check: $\cos(x) = x$ and $\cos^{-1}(x) = x$ have the same solution). Using the criterion from lectures for when a fixed-point iteration converges, explain why one of these settles down nicely and the other doesn't. You'll need $g'(x)$ for each case, evaluated at the fixed point itself – you can find that fixed point to enough accuracy just by looking at where your iterations for part (a) are heading.

Part 2: Newton-Raphson

(d) Take $f(x) = x^3 - x^2 - 1$ and $x_0 = 1$. Calculate x_1, x_2, x_3 by hand, to 3 decimal places.

(e) Now take $f(x) = -x^4 + x^2 + \frac{20}{81}$ and $x_0 = \frac{1}{3}$. Calculate x_1, x_2, x_3 by hand, keeping everything as exact fractions rather than decimals. This one behaves very differently – and f here is a perfectly ordinary polynomial, smooth everywhere, nothing like the sharp corner or blow-up you might expect a "badly behaved" function to have.

(f) One of these converges steadily; the other locks into a perfect two-step cycle instead, forever swapping between x_0 and $-x_0$ and never getting anywhere near an actual root of f (its real roots are near ± 1.098). Show algebraically why the sign flips *exactly* every step in the cycling case. (Hint: what kind of function is f – even, odd, or neither? What does that make f'? Use this to compare $f(-x_0)/f'(-x_0)$ with $f(x_0)/f'(x_0)$.)

Why we are doing this

Every method in this module has a range of problems where it works beautifully and a range where it quietly falls apart. The point of this question isn't the arithmetic – it's noticing that "diverges" isn't a mysterious failure, it's something you can see coming directly from properties of the functions involved, sometimes exactly the criteria from the lecture notes, sometimes just careful algebra, before you ever run any code.

Related past exam questions

If you'd like to see this style tested in a real exam, these are worth a look. MTH2014 is a new module, so there's no MTH2014 past paper yet – these are from MTH2005 ("Modelling: Theory and Practice"), the module MTH2014 replaces, taken from the University's official past-paper archive. The root-finding content is the same.

- **May 2023, Q1(a)** and **May 2025, Q1(a)** – almost exactly parts (a)-(c) of this sheet: two choices of $g(x)$ for the same equation, one convergent and one not, asking you to decide which and explain why.
- **May 2021, Q1(a)** – the same idea again, with an extra part asking you to show the two choices of $g(x)$ come from the same original equation.
- **May 2024, Q1(b)** – proves the general convergence criterion used in part (c) ($|g'(\xi)| < 1$) from the Mean Value Theorem, rather than just using it.
- **May 2020, Q1(a)** and **May 2026, Q1(a)** – the same style of Newton-Raphson computation as part (d), plus a discussion question on its merits against bisection, which is exactly the kind of thing part (f) is getting at, asked directly.